



Okan Efe
Öğretmen

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Sarıyer - İstanbul / Turkey

Profile

Ph.D. candidate, researcher, designer, and developer of consumer and enterprise XR experiences. A multidisciplinary, fast learner, creative, and passionate about visual design, ambitious to predict and shape the future by creating**.

Currently I am looking for a lab to collaborate with any of the following subjects under visiting researcher title.

- UX research studies through XR
- Integrating XR tools into product design process
- Integrating XR into domestic use cases (Kitchen)
- Designing UI for XR technology

Education

PhD, Design, Technology and Society — Koç University, İstanbul, Türkiye

2020–2026

KARMA Mixed Reality Lab & KUAR Research Center.

Funded by TÜBİTAK 2244 Industrial PhD Fellowship (Project No. 118C095, industry partner: Beko).

Dissertation: Exploring the Design Opportunities for Mediated Spatial Interactions Learning from the Kitchen

Supervisor: Assoc. Prof. Asım Evren Yantaç

MS, Information Technologies — FMV Işık University, İstanbul, Türkiye

2013–2014

With scholarship.

Game Art & Design — VanArts - Vancouver Institute of Media Arts, Vancouver, BC, Canada

2010–2011

With scholarship.

BS, Physics — FMV Işık University, İstanbul, Türkiye

2004–2009

With scholarship.

Publications

Öğretmen, O. E. Cılızoğlu, S. & Yantaç, A. E. (2026). **Augmented Kitchens: XR-Enabled Communication in Active Domestic Workspaces. In Extended Reality — XR Salento 2026**, Otranto, Italy. Springer LNCS. [DOI/pages when published]

Öğretmen, O. E., Akbaş, S., & Yantaç, A. E. (2025). **Exploring Designers' Expectations From a Virtual Reality User Insight Tool: KitchXR**. In Proceedings of the ACM Symposium on Spatial User Interaction (SUI '25). ACM. <https://doi.org/10.1145/3694907.3765914>

Kim, J., Nicenboim, I., Öğretmen, O. E., Martins, J., & Karana, E. (2026). **From machinery to sensibility: (Un)learning cyanobacteria between biological laboratory practice and more-than-human exploration**. BioDesign Conference 2026, TU Delft, Delft, Netherlands (peer-reviewed pictorial).

Akgün, M. Y., Baran, O., Öğretmen, E., Uçar, E. A., Öktenoğlu, T., Akyoldaş, G., Özer, A. F., & Ateş, Ö. (2024). **Use of 3-Dimensional Modeling and Virtual Reality in the Education of Posterior Spinal Instrumentation**. Turkish Neurosurgery, 958–965. <https://doi.org/10.5137/1019-5149.JTN.46204-24.1>

Öğretmen, O. E. Cılızoğlu, S. & Yantaç, **Hybrid Connected Kitchens: Phygital Social XR for Domestic Communication and Collaboration**. Under review at a peer-reviewed journal (Springer).

Experience

Part-Time Instructor — Kadir Has University, Istanbul, Türkiye 2026–present

NMD220 - Game Programming

Part-Time Instructor — Koç University, Istanbul, Türkiye 2025–present

Mava357 - Designing Extended Reality Experiences

Visiting Researcher — DREAM Lab, Faculty of Industrial Design Engineering, TU Delft, Netherlands Aug 2024 – Feb 2025

Host: Prof. Elvin Karana. Developed augmented and immersive tools for biodesign research, including the HMD based Cyano Vision passthrough tool (Unity, Meta Quest 3) contributed to the BioDesign 2026 pictorial.

Teaching Assistant — Koç University, Istanbul, Türkiye 2020–2026

- MAVA 357 Designing Extended Reality Experiences — delivered lectures on 3D modeling in Blender; mentored student groups through project development.
- MAVA 205 Creative Thinking — led a student group through the term.
- DSEG 531 Design for Extended Reality — consulted graduate project groups during development.

Mentorship— 2020–2026

- Meta XR Hackathon
- CoskunÖz XR Akademi
- Designing The Future With Gifted - VR Eğitimi - MEB Bilsem - Ülker Yörükoğlu

Part-Time Instructor — Graphic Design Department, FMV Işık University Vocational School, Istanbul 2017-2020

Teaching Assistant (Maya) — VanArts, Vancouver, BC, Canada 2011

Selected Projects

- **Hybrid Connected Kitchens — PhD research programme** 2020–2026 A four-study programmatic design research arc (Kitchen Talks, Portal, KitchXR, Hybrid Connected Kitchens) investigating XR-mediated communication in domestic spaces. Designed and conducted co-design studies, participant observation, and expert evaluations; built VR/MR prototypes in Unity (C#); evaluated with thematic analysis and the Continuous Fuzzy Kano Model.
 - **Cultural Heritage with XR — Vehbi Koç Vakfı & KARMA Mixed Reality Lab** 2024-2026 Creating virtual twins of archaeological sites and building immersive experiences with advanced storytelling. Photogrammetry, 3D modeling, and VR/MR experience development in Unity.
 - **Gordion Museum App — Android tablet application** 2024-2025 Presents a 3D virtual twin of the Gordion Tomb alongside images from the first excavation. Designed the UX and UI and built the entire application in Unity.
 - **Cyano Vision — TU Delft** 2024–2025 Speculative VR passthrough tool transforming camera data into a representation of cyanobacterial perception (Unity, Meta Quest 3); developed for more-than-human biodesign research.
 - **KUH Surgery VR — Koç University Hospital** 2022 VR spine and brain-surgery training environment for medical conferences. Level design, modeling and optimization, UI design, C# programming (Oculus Interaction Kit), shading, render optimization. Basis of the Turkish Neurosurgery publication.
 - **Knobie — MR menu-design prototype** 2021 HoloLens tool combining printed recipe cards with sustainability, seasonality, and price scores. Card and UI design; Unity/Vuforia development; HoloLens optimization.
 - **Dünyalı AR — educational AR application** 2021 XR consultant for a map and globe publisher: ideation, UX, 3D design, and development; published on App Store and Play Store.
 - **Hacivat & Karagöz Educational Game — MS thesis project** 2013 Educational game for pre-school children based on the traditional Turkish shadow-play characters. 2D character design, 3D character modeling, and gesture-based interaction design with Leap Motion.
 - **Hi-Menu — in-restaurant tablet menu & entertainment platform** 2012 Spin-off project delivering menus and customer entertainment on iPads in restaurants and cafés. Ideation, UX/UI design, front-end development, and user testing.
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Skills

XR Development

Unity (C#), Meta/Oculus SDK & Interaction SDK, OpenXR, SteamVR, Vuforia

3D

Blender, Autodesk Maya, ZBrush — modeling, texturing, shading, lighting

Design

UI/UX for XR, design thinking, co-design facilitation, Miro, Adobe CC (Photoshop, Illustrator, InDesign), Affinity Designer, desktop publishing

Research Methods

Research-through-Design, Programmatic Design Research, co-design studies, participant observation, thematic analysis, Continuous Fuzzy Kano Model, expert evaluation

Other

HTML/CSS, Python (data analysis), Agile/Scrum project management, photography & photogrammetry

Languages

Turkish (native) · English (fluent)

References

Available upon request.

Studies

- **Ethics of Avatar** (2023), I have designed and conducted a co-design study to start a discussion about the ethics of avatar design. During the study, Miro, online surveys, and offline materials are being used during a face-to-face group meeting. Participants have been guided to use AI tools for creating personas and avatars (thispersondoesnotexist.com and [mid-journey](https://mid-journey.com)). I completed designing study, designing materials, designing dynamic surveys, and analyzing the findings.
 - **Workathon** (2023), A co-design study for my Ph.D, conducted with designers from home appliances industry. Study consist of three stages, and expanded to one month study. Introduction, is the first gathering session held online, and after the introduction of participants, I presented the concept and briefed technologies to be discuss during the study. Ideation session held offline, a full day study, conducted with design thinking process. Third session, the hackathon, held offline for three days, participants developed the ideas and virtually prototyped them inside the VR Kitchen tool. With this study I aimed to discover gaps during the design process and opportunities on mediated spatial interaction design in kitchen. I have designed all the processes and materials to be used. These
 - **Kitchen Talks** (2022), Another co-design study with kitchen users, during my Ph.D. aims to find design opportunities for a communication tool from kitchen, consist of three main stages, warming session, gathering insights and brainstorming. Study conducted online through Miro and zoom. I completed the designing study, designing materials, conducting study and analyzing the findings (with thematic analysis techniques) processes.
 - **User Studies of VR Kitchen** (2021), I conducted user studies for the VR Kitchen study. During the study, I observed and participants while experiencing the prototype and conducted a semi-instructed interview.
 - **Participant Observation** (2020), As part of my Ph.D studies, I participated online cooking workshops to conduct participant observation study. These workshops are conducted by a local non-profit initiative, with aiming cooking together vegetarian local recipes and expanding communication. I aimed to gather insights during cooking sessions and workshops. This study guided me to focus on communication from kitchen and Kitchen Talks , co-design study.
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